

Project Design Phase-II

Data Flow Diagram & User Stories

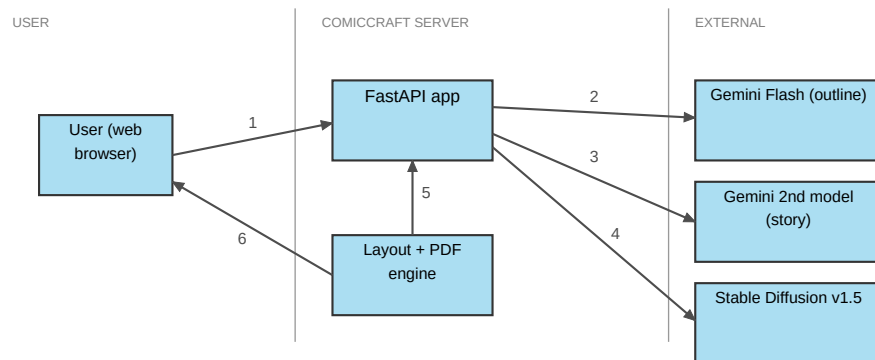
Date	2 October 2026
Team ID	12
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

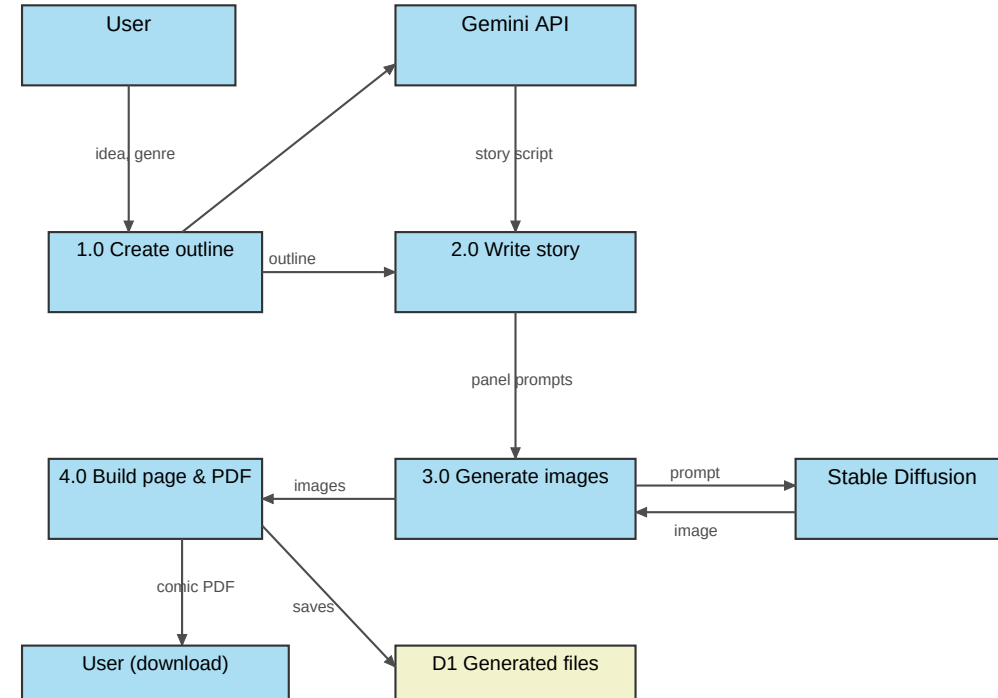
Example: [\(Simplified\)](#)

Flow



1. User enters a story idea, genre and panel count in the web UI.
2. FastAPI sends the idea to Gemini Flash to create an outline.
3. The outline goes to the second Gemini model, which writes the panel story.
4. Each panel scene is turned into a prompt for Stable Diffusion v1.5 (MPS, fp16).
5. The layout engine places images and captions on the comic page.
6. The page is shown in the browser and exported as a PDF with fpdf2.

Example: DFD Level 0 (Industry Standard) - ComicCraft



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Creator	Idea input	USN-1	As a user, I can type a one-line story idea on the home page.	Idea is accepted and validated	High	Sprint-1
		USN-2	As a user, I can choose a genre and panel count.	Options are saved with the request	High	Sprint-1
	Story generation	USN-3	As a user, I get a story outline from Gemini Flash.	Outline with title and panels is returned	High	Sprint-1
		USN-4	As a user, I get a full story script split into panels.	One caption per panel is produced	High	Sprint-2
	Image generation	USN-5	As a user, I get an image for every panel.	Image created by Stable Diffusion v1.5	High	Sprint-2
		USN-6	As a user, I can see progress while panels render.	Status updates until all panels finish	Medium	Sprint-2
	Comic layout	USN-7	As a user, I see the panels arranged on a comic page.	Grid layout with captions in each panel	High	Sprint-3
		USN-8	As a user, I want captions to fit inside their panels.	No caption text overflows a panel	Medium	Sprint-3
	PDF export	USN-9	As a user, I can download my comic as a PDF.	PDF opens with all panels and captions	High	Sprint-3
		USN-10	As a user, I can open and print the PDF offline.	PDF prints on A4 without cropping	Medium	Sprint-3
	Reliability	USN-11	As a user, I see a clear message when API limits are hit.	Friendly error with a retry option	Medium	Sprint-4
		USN-12	As a user, I can regenerate a comic with a changed idea.	New outline, story and images are produced	Low	Sprint-4